

```
!ls
```

```
sample_data
```

```
%mkdir test
```

```
!ls
```

```
sample_data  test
```

```
%cd test
```

```
/content/test
```

```
!ls
```

```
!pwd
```

```
/content/test
```

```
%%writefile /content/test/spacex_ships.py  
import requests
```

```
Writing /content/test/spacex_ships.py
```

```
%%writefile /content/test/spacex_ships.py  
  
import requests  
import json  
  
def fetch_data(endpoint):  
    url = f"https://api.spacexdata.com/v3/{endpoint}"  
    response = requests.get(url)  
    if response.status_code == 200:  
        return response.json()  
    else:  
        print(f"Failed to fetch data from {endpoint}")  
        return []  
  
def print_data(data):  
    print(json.dumps(data, indent=4))  
  
if __name__ == "__main__":  
    endpoint = input("Enter the SpaceX API endpoint (e.g., ships, launches, rockets): ")  
    data = fetch_data(endpoint)  
    print_data(data)
```

```
Overwriting /content/test/spacex_ships.py
```

```
!python spacex_ships.py
```

```
Enter the SpaceX API endpoint (e.g., ships, launches, rockets): ships  
[  
  {  
    "ship_id": "AMERICANCHAMPION",  
    "ship_name": "American Champion",  
    "ship_model": null,  
    "ship_type": "Tug",  
    "roles": [  
      "Support Ship",  
      "Barge Tug"  
    ],  
    "active": false,  
    "imo": 7434016,  
    "mmsi": 367020820,  
    "abs": 571252,  
    "class": 7604342,  
  },  
]
```

```

    "weight_lbs": 588000,
    "weight_kg": 266712,
    "year_built": 1976,
    "home_port": "Port of Los Angeles",
    "status": "",
    "speed_kn": null,
    "course_deg": null,
    "position": {
        "latitude": null,
        "longitude": null
    },
    "successful_landings": null,
    "attempted_landings": null,
    "missions": [
        {
            "name": "COTS 1",
            "flight": 7
        },
        {
            "name": "COTS 2",
            "flight": 8
        }
    ],
    "url": "https://www.marinetraffic.com/en/ais/details/ships/shipid:434663/vessel:AMERICAN%20CHAMPION",
    "image": "https://i.imgur.com/woCxpKj.jpg"
},
{
    "ship_id": "AMERICANISLANDER",
    "ship_name": "American Islander",
    "ship_model": null,
    "ship_type": "Cargo",
    "roles": [
        "Dragon Recovery"
    ],
    "active": false,
    "imo": null,
    "mmsi": 367035570,
    "abs": null,
    "class": null,
    "weight_lbs": null,
    "weight_kg": null
}

```

```

# task 3.1
from datetime import datetime
print(datetime.now())

```

2026-05-28 19:22:44.464480

```

# task 3.2
import re
pattern = re.compile(r'\d+')
print(pattern.findall('There are 123 numbers in 456 this string.'))

```

['123', '456']

```

# task 3.3
import json
data = '{"name": "John", "age": 30}'
parsed = json.loads(data)
print(parsed['name'])

```

John

```

# task 3.4
import math
print(math.pi)
print(math.sqrt(16))

```

3.141592653589793
4.0

```

# task 3.5
import random
print(random.randint(1, 100))

```

51

```
# task 3.6
import itertools
for i in itertools.count(10):
    print(i)
    if i >= 15:
        break
```

```
10
11
12
13
14
15
```

```
# task 3.7
import os
print(os.name)
print(os.listdir('.'))
```

```
posix
['spacex_ships.py']
```

```
# task 3.8
import sys
print(sys.platform)
print(sys.version)
```

```
linux
3.12.13 (main, Mar  4 2026, 09:23:07) [GCC 11.4.0]
```

```
# task 3.9
from collections import Counter
print(Counter('hello world'))
```

```
Counter({'l': 3, 'o': 2, 'h': 1, 'e': 1, ' ': 1, 'w': 1, 'r': 1, 'd': 1})
```

```
# task 3.10
import functools
print(functools.reduce(lambda x, y: x + y, [1, 2, 3, 4, 5]))
```

```
15
```

```
# task 3.11
import operator
print(operator.add(1, 2))
```

```
3
```

```
# task 3.12
import subprocess
print(subprocess.run(['echo', 'Hello World!'], capture_output=True, text=True).stdout)
```

```
Hello World!
```

```
# task 3.13
import csv
with open('example.csv', mode='w', newline='') as file:
    writer = csv.writer(file)
    writer.writerow(['name', 'age'])
    writer.writerow(['Alice', 30])
    writer.writerow(['Bob', 25])
    with open('example.csv', mode='r') as file:
        reader = csv.reader(file)
        for row in reader:
            print(row)
```

```
# task 3.14
import shutil
shutil.copyfile('example.csv', 'example_copy.csv')
```

```
'example_copy.csv'
```

```
# task 3.15
```

```
from pathlib import Path

path = Path('example.csv')
print(path.exists())
print(path.is_file())
```

```
True
True
```

```
# task 3.16
import requests
response = requests.get('https://api.github.com')
print(response.status_code)
print(response.json())
```

```
200
{'current_user_url': 'https://api.github.com/user', 'current_user_authorizations_html_url': 'https://github.com/setti
```

```
# task 3.17
from bs4 import BeautifulSoup
html = '<html><head><title>Test</title></head><body><p>Hello, world!</p></body></html>'
soup = BeautifulSoup(html, 'html.parser')
print(soup.title.string)
print(soup.p.text)
```

```
Test
Hello, world!
```

```
# task 3.18
import numpy as np
array = np.array([1, 2, 3, 4])
print(array * 2)
```

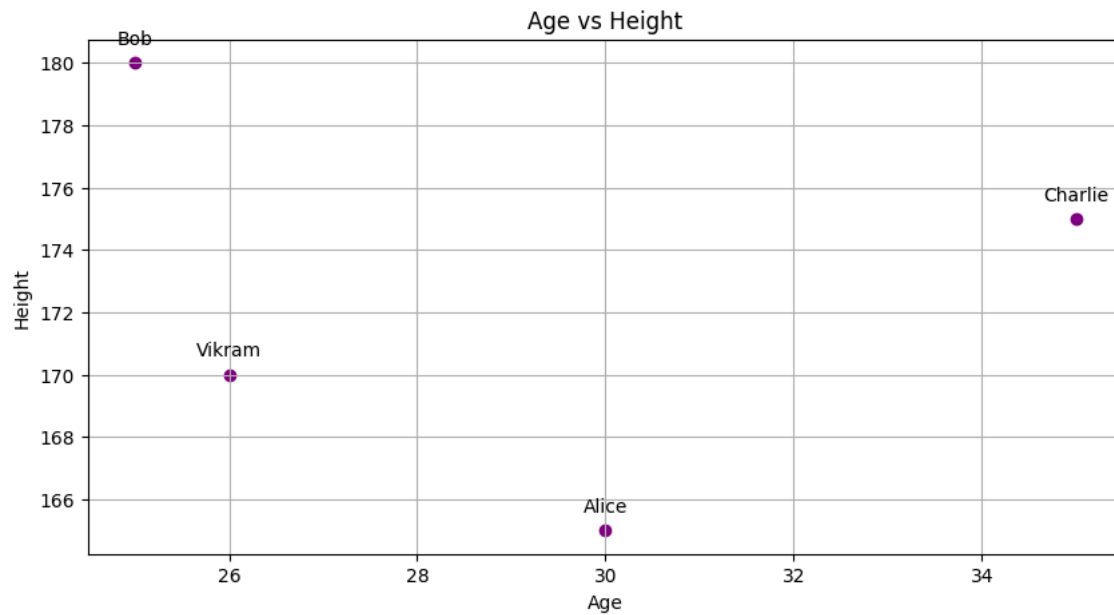
```
[2 4 6 8]
```

```
# task 3.19
#add your name age can be fake
import pandas as pd
data = {'name': ['Alice', 'Bob', 'Charlie', 'Vikram'], 'age': [30, 25, 35, 26]}
df = pd.DataFrame(data)
print(df.head())
print(df.describe())
print(df[df['age'] > 30])
df['height'] = [165, 180, 175, 170]
print(df)
```

```
   name  age
0  Alice   30
1   Bob   25
2  Charlie  35
3  Vikram   26
age
count    4.000000
mean    29.000000
std      4.546061
min     25.000000
25%     25.750000
50%     28.000000
75%     31.250000
max     35.000000
name  age
2  Charlie  35
name  age  height
0  Alice   30    165
1   Bob   25    180
2  Charlie  35    175
3  Vikram   26    170
```

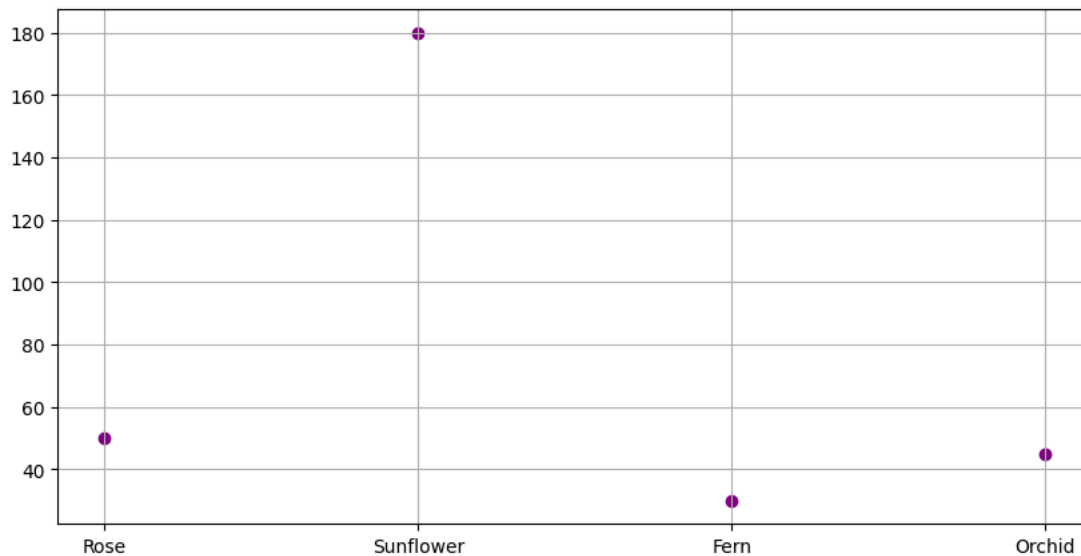
```
# task 3.20
#this code is continuation of 3.19
import matplotlib.pyplot as plt
plt.figure(figsize=(10, 5))
plt.scatter(df['age'], df['height'], color='purple')
for i, name in enumerate(df['name']):
    plt.annotate(name, (df['age'][i], df['height'][i]), textcoords="offset points", xytext=(0,10), ha='center')
plt.xlabel('Age')
```

```
plt.ylabel('Height')
plt.title('Age vs Height')
plt.grid(True)
plt.show()
```



```
# VN - my example
import matplotlib.pyplot as plt
import pandas as pd
data = {'plant_name': ['Rose', 'Sunflower', 'Fern', 'Orchid'], 'height_cm': [50, 180, 30, 45]}
df = pd.DataFrame(data)
df['leave_size'] = [75, 150, 50, 55]
plt.figure(figsize=(10, 5))
plt.grid(True)
plt.scatter(df['plant_name'], df['height_cm'], color='purple')
```

<matplotlib.collections.PathCollection at 0x7ec3c938b3b0>



```
# Write the module
%writefile myutils/mathutils.py
def add(a, b):
    return a + b
def subtract(a, b):
    return a - b
def multiply(a, b):
    return a * b
def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
```

```
return a / b
```

Overwriting myutils/mathutils.py

```
from google.colab import drive
drive.mount('/content/drive')
```

```
# Write the unit tests
%%writefile myutils/tests/test_mathutils.py
import unittest
from myutils.mathutils import add, subtract, multiply, divide
class TestMathUtils(unittest.TestCase):
    def test_add(self):
        self.assertEqual(add(2, 3), 5)
        self.assertEqual(add(-1, 1), 0)
        self.assertEqual(add(-1, -1), -2)
    def test_subtract(self):
        self.assertEqual(subtract(2, 3), -1)
        self.assertEqual(subtract(-1, 1), -2)
        self.assertEqual(subtract(-1, -1), 0)
    def test_multiply(self):
        self.assertEqual(multiply(2, 3), 6)
        self.assertEqual(multiply(-1, 1), -1)
        self.assertEqual(multiply(-1, -1), 1)
    def test_divide(self):
        self.assertEqual(divide(6, 3), 2)
        self.assertEqual(divide(-1, 1), -1)
        self.assertEqual(divide(-1, -1), 1)
        with self.assertRaises(ValueError):
            divide(1, 0)
if __name__ == '__main__':
    unittest.main(argv=[''], verbosity=2, exit=False)
```

Overwriting myutils/tests/test_mathutils.py